

# SDLC V-Model - Software Engineering

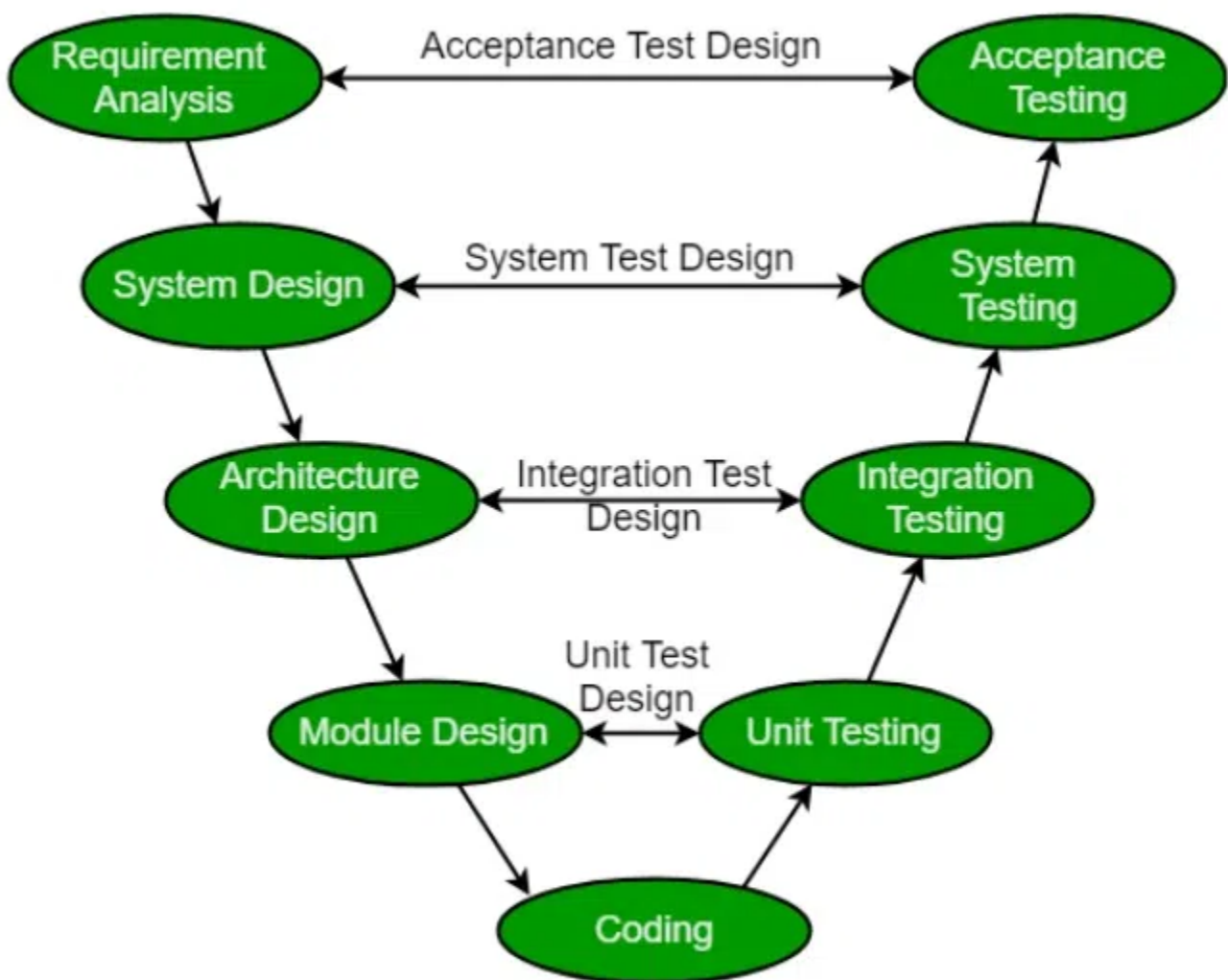
Last Updated : 11 Aug, 2025



The SDLC V-Model is a type of Software Development Life Cycle (SDLC). It is a method that includes testing and validation alongside each development phase. It creates a structure like the letter 'V,' which includes various phases that we will discuss in detail.

## Phases of SDLC V-Model

The V-Model, which includes the Verification and Validation it is a structural approach to software development. The following are the different Phases of the V-Model of the SDLC.



# 1. V-Model Verification Phases

This is where the process begins. The first step is to gather and understand the customer's needs for the software. The goal is to define the scope of the project clearly to make sure everyone is on the same page.

It involves a static analysis technique (review) done without executing code. It is the process of evaluation of the product development phase to find whether specified requirements are met.

There are several Verification phases in the V-Model:

## 1. Business Requirement Analysis

This is the first step of the designation of the development cycle where product requirement needs to be cured from the customer's perspective. These phases include proper communication with the customer to understand their requirements.

These are the very important activities that need to be handled properly, as most of the time customers do not know exactly what they want, and they are not sure about it at that time then we use an **acceptance test design** planning which is done at the time of business requirement it will be used as an **input** for acceptance testing.

## 2. System Design

In this phase, the overall structure of the software is planned out. The team develops both the high-level design (how the system will be structured) and detailed design (how the individual components will work).

Design of the system will start when the overall we are clear with the product requirements, and then need to design the system completely. This understanding will be at the beginning of complete under the product development process. these will be beneficial for the future execution of test cases.

## 3. Architectural Design

In this stage, architectural specifications are comprehended and designed. Usually, several technical approaches are put out, and the ultimate choice is made after considering both the technical and financial viability. The system architecture is further divided into modules that each handle a distinct function. Another name for this is High-Level Design (HLD).

At this point, the exchange of data and communication between the internal modules and external systems are well understood and defined. During this phase, integration tests can be created and documented using the information provided.

## 4. Module Design

This phase, known as Low-Level Design (LLD), specifies the comprehensive internal design for every system module. Compatibility between the design and other external systems as well as other modules in the system architecture is crucial. Unit tests are a crucial component of any development process since they assist in identifying and eradicating the majority of mistakes and flaws at an early stage. Based on the internal module designs, these unit tests may now be created.

## 5. Coding Phase

This is where the software is actually built. Developers write the code based on the design created in the previous phase.

The Coding step involves writing the code for the system modules that were created during the Design phase. The system and architectural requirements are used to determine which programming language is most appropriate.

The coding standards and principles are followed when performing the coding. Before the final build is checked into the repository, the code undergoes many code reviews and is optimized for optimal performance.

## 2. V-Model Validation Phases

It involves dynamic analysis techniques (functional, and non-functional), and testing done by executing code. Validation is the process of evaluating the

software after the completion of the development phase to determine whether the software meets the customer's expectations and requirements.

## 1. Unit Testing

In [Unit testing](#), unit Test Plans are developed during the module design phase. These Unit Test Plans are executed to eliminate bugs in code or unit level.

## 2. Integration testing

After completion of unit testing [Integration testing](#) is performed. In integration testing, the modules are integrated and the system is tested. Integration testing is performed in the Architecture design phase. This test verifies the communication of modules among themselves.

## 3. System Testing

[System testing](#) tests the complete application with its functionality, inter-dependency, and communication. It tests the functional and non-functional requirements of the developed application.

## 4. User Acceptance Testing (UAT)

[User Acceptance Testing \(UAT\)](#) is performed in a user environment that resembles the production environment. UAT verifies that the delivered system meets the user's requirement and the system is ready for use in the real world.

## Industrial Challenge of V-Model

As the industry has evolved, the technologies have become more complex, increasingly faster, and forever changing, however, there remains a set of basic principles and concepts that are as applicable today as when IT was in its infancy.

- Accurately define and refine user requirements.
- Design and build an application according to the authorized user requirements.
- Validate that the application they had built adhered to the authorized business requirements.

# Importance of V-Model

The V-Model is an important part of the SDLC, and the process is structured and sequential throughout all the testing. Here is why the V-Model is important:

## 1. Early Defect Identification

By incorporating verification and validation tasks into every stage of the development process, the V-Model encourages early testing. This lowers the cost and effort needed to remedy problems later in the development lifecycle by assisting in the early detection and resolution of faults.

## 2. Determining the Phases of Development and Testing

The V-Model contains a testing phase that corresponds to each stage of the development process. By ensuring that testing and development processes are clearly mapped out, this clear mapping promotes a methodical and orderly approach to software engineering.

## 3. Prevents "Big Bang" Testing

Testing is frequently done at the very end of the development lifecycle in traditional development models, which results in a "Big Bang" approach where all testing operations are focused at once. By integrating testing activities into the development process and encouraging a more progressive and regulated testing approach, the V-Model prevents this.

## 4. Improves Cooperation

At every level, the V-Model promotes cooperation between the testing and development teams. Through this collaboration, project requirements, design choices, and testing methodologies are better understood, which improves the effectiveness and efficiency of the development process.

## 5. Improved Quality Assurance

Overall quality assurance is enhanced by the V-Model, which incorporates testing operations at every level. Before the program reaches the final deployment stage, it makes sure that it satisfies the requirements and goes through a strict validation and verification process.

## Principles of V-Model

- **Large to Small:** In V-Model, testing is done in a hierarchical perspective, for example, requirements identified by the project team, creating High-Level Design, and Detailed Design phases of the project. As each of these phases is completed the requirements, they are defining become more and more refined and detailed.
- **Data/Process Integrity:** This principle states that the successful design of any project requires the incorporation and cohesion of both data and processes. Process elements must be identified at every requirement.
- **Scalability:** This principle states that the V-Model concept has the flexibility to accommodate any IT project irrespective of its size, complexity, or duration.
- **Cross Referencing:** A direct correlation between requirements and corresponding testing activity is known as cross-referencing.

## Tangible Documentation:

This principle states that every project needs to create a document. This documentation is required and applied by both the project development team and the support team. Documentation is used to maintain the application once it is available in a production environment.

## Why preferred?

- It is easy to manage due to the rigidity of the model. Each phase of V-Model has specific deliverables and a review process.
- Proactive defect tracking – that is defects are found at an early stage.

## When to Use of V-Model?



- **Traceability of Requirements:** The V-Model proves beneficial in situations when it's imperative to create precise traceability between the requirements and their related test cases.
- **Complex Projects:** The V-Model offers a methodical way to manage testing activities and reduce risks related to integration and interface problems for projects with a high level of complexity and interdependencies among system components.
- **Waterfall-Like Projects:** Since the V-Model offers an approachable structure for organizing, carrying out, and monitoring testing activities at every level of development, it is appropriate for projects that use a sequential approach to development, much like the waterfall model.
- **Safety-Critical Systems:** These systems are used in the aerospace, automotive, and healthcare industries. They place a strong emphasis on rigid verification and validation procedures, which help to guarantee that essential system requirements are fulfilled and that possible risks are found and eliminated early in the development process.

## Advantages of V-Model

- This is a highly disciplined model and Phases are completed one at a time.
- V-Model is used for small projects where project requirements are clear.
- Simple and easy to understand and use.
- This model focuses on verification and validation activities early in the life cycle thereby enhancing the probability of building an error-free and good quality product.
- It enables project management to track progress accurately.
- The V-Model provides a clear and structured process for software development, making it easier to understand and follow.
- The V-Model places a strong emphasis on testing, which helps to ensure the quality and reliability of the software.
- The V-Model provides a clear link between the requirements and the final product, making it easier to trace and manage changes to the software.
- The clear structure of the V-Model helps to improve communication between the customer and the development team.

## Disadvantages of the V-Model

- The V-Model is a linear and sequential model, which can make it challenging to adapt to changing requirements or unforeseen events.
- The V-Model can be time-consuming, as it requires a lot of documentation and testing.
- High risk and uncertainty.
- It is not good for complex and object-oriented projects.
- It is not suitable for projects where requirements are not clear and contain a high risk of changing.
- This model does not support iteration of phases.
- The V-Model places a strong emphasis on documentation, which can lead to an overreliance on documentation at the expense of actual development work.

The SDLC V-Model is a effective framework for project completion with suitable process which is included in the Software Development Life Cycle (SDLC) you can read more about the all models which is "[Software Development Life Cycle \(SDLC\) Models](#)".

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